FIELD NOTES · NO. 01

HOW DNS RESOLUTION WORKS

Notes by Ruben de Beer

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I

OVERVIEW

Every time you type a name into a browser, a small relay race runs across the internet to turn it into an address.

DNS is the internet's phone book, except nobody holds the whole book. Each server knows one small part of the tree and who to ask about the rest. Your machine asks one resolver; that resolver does the walking for you and remembers the answer for as long as the record's *time to live* allows.

13	53	512
ROOT SERVER NAMES	PORT, UDP & TCP	BYTES, CLASSIC UDP LIMIT

What actually happens

- Your stub resolver sends the question to a recursive resolver.
- The recursive resolver answers from cache, or walks root → TLD → authoritative.
- The answer comes back with a TTL, and everyone along the way may cache it.

SERVER	KNOWS ABOUT	TYPICAL ANSWER
Root	Top-level domains	A referral to the .com servers
TLD	Domains under it	A referral to the domain's nameservers
Authoritative	One zone's records	The record itself: A, AAAA, MX…

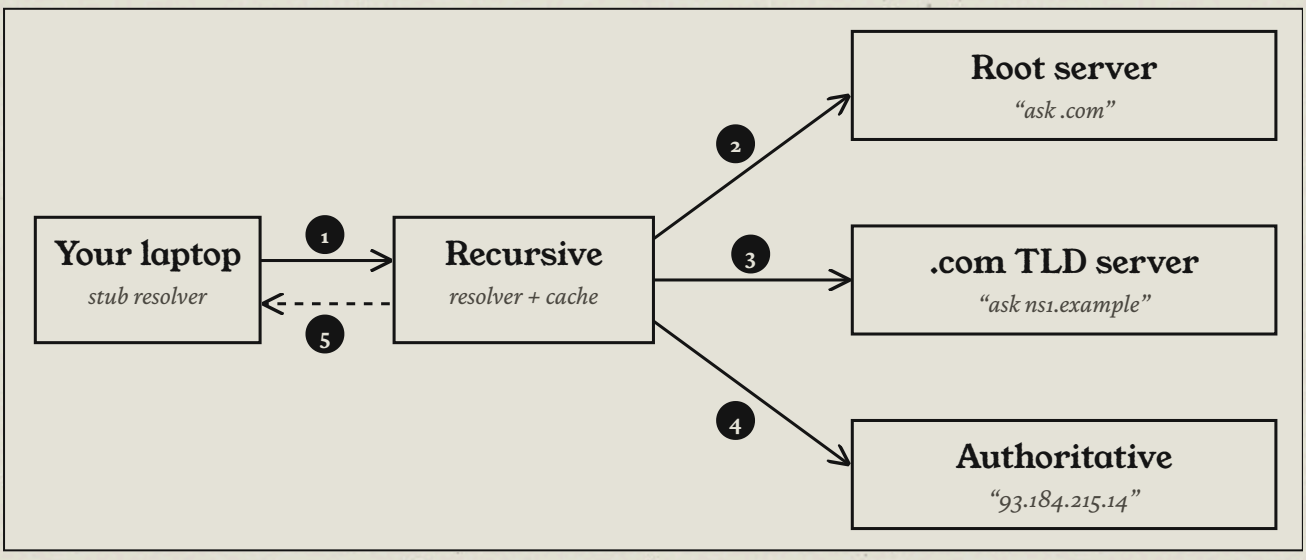
> **REMEMBER**
> Most lookups never leave the first resolver: a warm cache answers in a couple of milliseconds.

↳ this is why "it works on my machine" can be a DNS cache

II

THE LOOKUP

One question, up to four hops, then a long nap in a cache.



A COLD LOOKUP FOR EXAMPLE.COM

Steps 2–4 only happen on a cache miss

RECURSIVE RESOLVER

Does the legwork for you: follows referrals, validates DNSSEC if asked, and caches every answer it sees.

AUTHORITATIVE SERVER

Holds the zone file. It only answers for names it owns, and says so with the `AA` flag.

REFERRAL

"I don't know, but these servers do." Roots and TLDs mostly answer this way.

TTL

How long, in seconds, an answer may be reused. Short TTLs make changes fast and caches busy.

"It's always DNS."

EVERY ON-CALL ENGINEER, EVENTUALLY

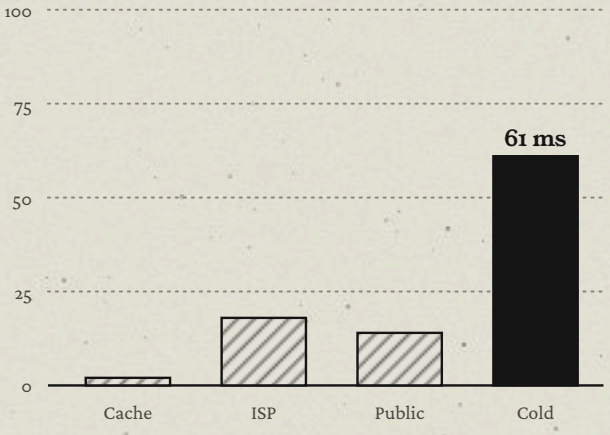
DEBUGGING TIP

`dig +trace example.com` walks the whole chain from the root and prints every referral.

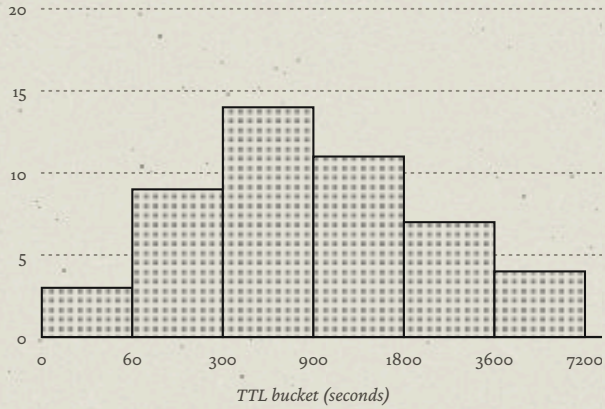
III

BY THE NUMBERS

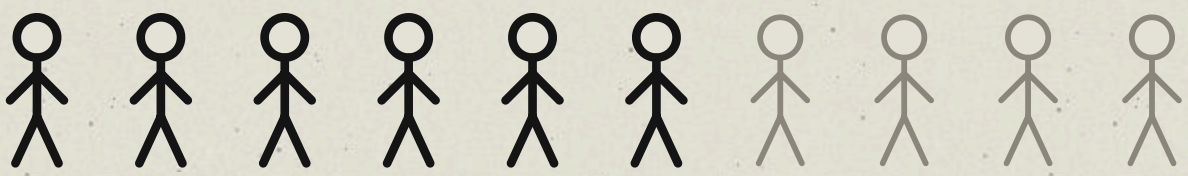
Illustrative figures for a small office network. Sample data, not measurements.



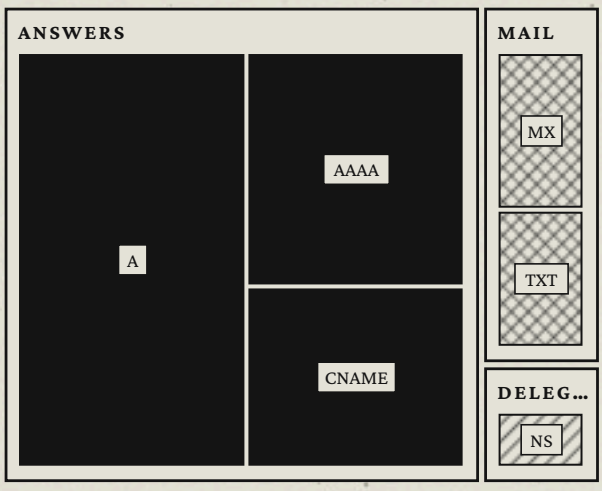
LOOKUP TIME BY PATH *sample data*



RECORDS BY TTL *sample data*



6 IN 10 LOOKUPS ANSWERED FROM CACHE *sample data*



RECORDS IN ONE ZONE *sample*

READING THE CHARTS

Charts print in ink only. Patterns replace colours: solid for the series you want seen first, then cross-hatch, hatch, dots and plain paper.

Answers · Mail · Delegation

IV

RECORDS

The record types you meet most, and what each one is for.



SHARE OF RECORDS IN ONE ZONE — *sample data*

TYPE	HOLDS	EXAMPLE VALUE
A	An IPv4 address	`93.184.215.14`
AAAA	An IPv6 address	`2606:2800:21f:cb07::1`
CNAME	An alias to another name	`www → example.com`
MX	Mail servers, with priority	`10 mail.example.com`
TXT	Free text: SPF, verification	`v=spf1 -all`
NS	The zone's nameservers	`ns1.example.com`

• • •

i. A CNAME can't sit at the zone apex next to other records.

ii. Lower TTLs a day before a migration, not an hour before.

iii. Negative answers (NXDOMAIN) are cached too.

NOTE

Shaded callouts sit one step darker than the page, for asides that shouldn't shout.

Until the next edition,

Ruben

Notes · Opinions · Contact rubendebeer.github.io/Rubi-Tubes

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